Improving Crop Disease Detection Using Multi-Dataset Training and Domain Generalization

Final Technical Report – Deliverable 3

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Track C – Research Paper Implementation

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Project Title: Improving Crop Disease Detection Using Multi-Dataset Training and Domain Generalization

Selected Research Paper:
M. S. Krishna, P. Machado, R. I. Otuka, S. W. Yahaya, F. Neves dos Santos, and I. K. Ihianle, "Plant Leaf Disease Detection Using Deep Learning: A Multi-Dataset Approach," *J*, vol. 8, no. 1, p. 4, 2025.

Contents

1 Abstract

Crop diseases remain a major threat to agricultural productivity, crop quality, farmer income, and food security. The problem is particularly significant because traditional crop disease diagnosis is often carried out manually through visual inspection by farmers or domain experts, making the process time-consuming, subjective, and difficult to scale. Deep learning has shown considerable promise for image-based plant disease detection; however, many state-of-the-art systems are trained on controlled datasets and fail to generalize adequately in real agricultural environments. This limitation is primarily caused by domain shift, where deployment images differ substantially from training images in terms of lighting, background clutter, image quality, camera angle, and symptom visibility.

This project presents a full implementation and analysis of the research paper *Plant Leaf Disease Detection Using Deep Learning: A Multi-Dataset Approach*. The project adopts a multi-dataset strategy by combining PlantDoc and a web-sourced plant disease dataset in order to improve robustness under diverse image conditions. The complete technical workflow included data acquisition, cleaning, preprocessing, exploratory data analysis, model implementation, training, evaluation, comparative analysis, and interpretation of failure cases. The system was implemented in PyTorch and evaluated using six convolutional neural network architectures: MobileNetV2, ResNet18, EfficientNet-B0, ResNet50, DenseNet201, and EfficientNet-B3.

The final experimental results show that DenseNet201 achieved the strongest overall performance, with an accuracy of 73.77%, weighted precision of 0.7491, weighted recall of 0.7377, weighted F1-score of 0.7360, and AUC-ROC of 0.9678. MobileNetV2 produced the second-best overall results despite being significantly lighter than the heavier architectures, which demonstrates that lightweight models can remain competitive when trained on sufficiently diverse data. ResNet50 and EfficientNet-B0 produced stable and competitive results, while EfficientNet-B3 did not outperform EfficientNet-B0 in this implementation. Across all models, the most difficult disease categories were visually similar tomato leaf diseases, particularly tomato bacterial spot, tomato early blight, and tomato septoria leaf spot.

The findings support the main premise of the selected paper: multi-dataset training improves the realism and robustness of crop disease recognition by exposing CNNs to a broader image distribution. At the same time, the study highlights that important challenges remain, especially with respect to class imbalance, domain shift, and fine-grained inter-class similarity. This report provides a complete technical account of the implemented system, final results, model-wise analysis, limitations, and future directions.

2 Introduction

Agriculture is one of the most important sectors of the global economy because it supports food production, employment, trade, and rural livelihoods. Plant diseases have a direct negative impact on agricultural productivity by reducing yield quantity, reducing crop quality, and increasing the cost of treatment and management. In severe cases, disease outbreaks can spread rapidly and cause large-scale economic damage. The selected paper emphasizes that crop diseases are responsible for substantial yield losses worldwide, which makes timely and accurate disease diagnosis an essential agricultural problem.

Traditional disease detection typically depends on visual inspection by human experts. Although expert diagnosis can be effective, it is labour-intensive, slow, and vulnerable to inconsistency. Different diseases often share overlapping symptoms, which makes manual classification difficult, especially for non-experts. These limitations create a strong motivation for intelligent automated systems that can assist in disease detection from plant images.

Deep learning, particularly convolutional neural networks, has become a widely adopted solution for image-based disease detection because CNNs can learn hierarchical visual patterns directly from raw images. In the context of crop disease detection, CNNs can extract features related to lesion texture, color distribution, mildew formation, rust patterns, and morphological changes on leaves. However, many existing models are trained on clean benchmark datasets and do not perform equally well on real-world field images.

A key reason for this failure is domain shift. A model trained on clean or laboratory-style images may struggle when tested on images containing cluttered backgrounds, inconsistent illumination, blur, partial occlusion, or different camera angles. Addressing this challenge requires more diverse training data and more robust experimental design.

This project follows the methodology of the paper *Plant Leaf Disease Detection Using Deep Learning: A Multi-Dataset Approach*, which combines PlantDoc with a web-sourced dataset in order to improve data diversity and generalization. The selected paper explicitly frames its architecture around preprocessing, augmentation, model training, and evaluation, and this overall modular design was adopted in the present implementation. The paper also evaluates multiple CNN backbones, including ResNet50, DenseNet201, EfficientNet-B0, and EfficientNet-B3, which motivated the comparative study carried out in this report.

3 Research Paper Basis

The project is based on the following paper:

> M. S. Krishna, P. Machado, R. I. Otuka, S. W. Yahaya, F. Neves dos Santos, and I. K. Ihianle, "Plant Leaf Disease Detection Using Deep Learning: A Multi-Dataset Approach," *J*, vol. 8, no. 1, p. 4, 2025.

The selected paper makes several contributions that directly motivated this implementation:

- It proposes a multi-dataset strategy rather than relying on a single dataset.
- It combines PlantDoc with a web-sourced dataset to improve visual diversity.
- It evaluates multiple CNN architectures, including ResNet50, DenseNet201, EfficientNet-B0, and EfficientNet-B3.
- It emphasizes robust preprocessing, augmentation, and evaluation.

These aspects shaped the current project design. The implementation adopted the same central principle of combining multiple datasets, resizing images to 224×224, and using ImageNet normalization and augmentation. The architecture comparison in this report also extends the paper's spirit by including two additional baselines, MobileNetV2 and ResNet18, alongside the heavier paper-aligned models.

4 Problem Statement

The main problem addressed in this work is the development of a crop disease detection system that performs reliably across different image domains. While image-based deep learning systems for plant disease recognition have shown promise, many models fail when exposed to real-world images that differ from the visual conditions present in training data.

The central question of this project is:

> Can crop disease classification performance be improved by training CNN models on a combined multi-dataset image collection instead of relying on a single source dataset?

A secondary goal is to determine which CNN architecture performs best under this multi-dataset setting, and why certain architectures succeed or fail on specific disease classes.

5 Objectives

The main objectives of the project are:

1. To implement the selected research paper as a Track C research paper implementation.
2. To construct a multi-dataset crop disease classification pipeline using PlantDoc and a web-sourced dataset.
3. To clean, preprocess, and standardize the final image collection.
4. To perform exploratory data analysis and validate the class structure of the dataset.
5. To train and evaluate multiple CNN models under a common experimental setup.
6. To compare baseline and heavier architectures using accuracy, precision, recall, F1-score, and AUC-ROC.
7. To identify difficult disease categories and analyze the reasons for failure.
8. To produce a production-ready GitHub repository and a final technical report.

6 Dataset Description

6.1 Datasets Used

Two datasets were used in this implementation:

- **PlantDoc**: a real-world plant disease image dataset containing crop disease images under natural conditions.
- **Web-Sourced Dataset**: a disease image dataset associated with the selected paper and collected from online sources.

These datasets were merged to create a combined dataset with broader visual diversity. The aim of this design was to reduce overfitting to a single visual domain and improve disease recognition under varying environmental conditions.

6.2 Target Classes

The final implementation used 16 classes:

1. apple_leaf_healthy
2. apple_leaf_rust
3. apple_leaf_scab
4. corn_grey_leaf_spot
5. corn_leaf_blight
6. corn_leaf_healthy
7. corn_leaf_rust
8. potato_leaf_blight
9. potato_leafroll_virus
10. tomato_leaf_bacterial_spot
11. tomato_leaf_early_blight
12. tomato_leaf_healthy
13. tomato_leaf_late_blight
14. tomato_leaf_mould
15. tomato_leaf_powdery_mildew
16. tomato_septoria_leaf_spot

6.3 Validation Strategy

The final combined dataset was split as follows:

- Training set: 70%
- Validation set: 15%
- Test set: 15%

This split was selected to allow training, tuning, and unbiased final evaluation.

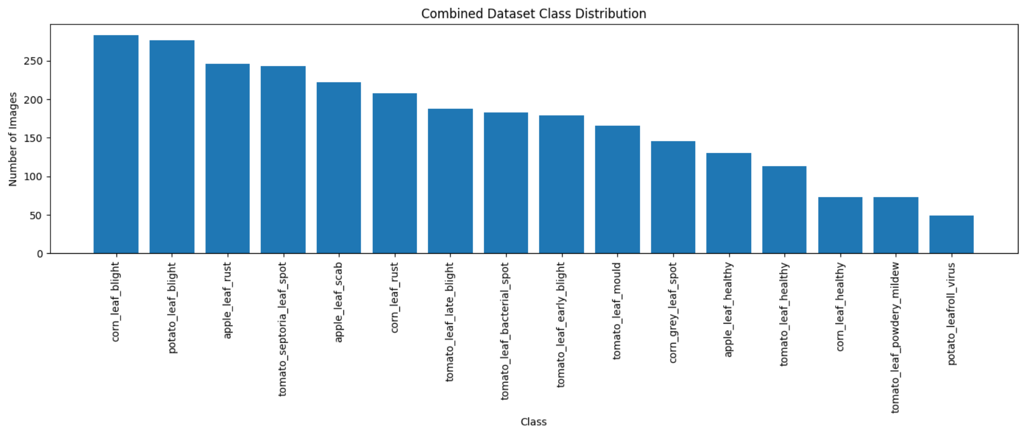

Figure 1: Combined dataset class distribution.

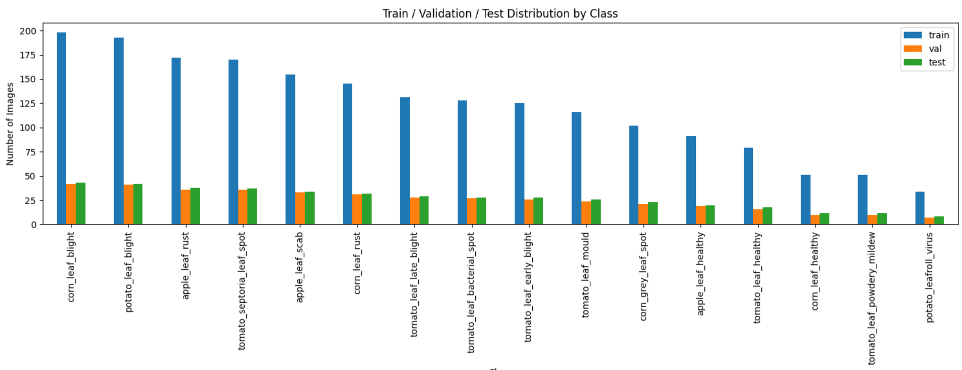

Figure 2: Train, validation, and test split distribution across all classes.

7 Data Cleaning and Preprocessing

The dataset underwent a structured cleaning and preprocessing pipeline before training.

7.1 Data Cleaning

The cleaning stage included:

- duplicate image removal,
- corrupted file removal,
- filtering of extremely small or unusable images.

This helped ensure that the final training set was not affected by repeated or invalid samples.

7.2 Preprocessing

All images were resized to 224×224 pixels and normalized using ImageNet statistics. These preprocessing steps were selected because they match the input assumptions of the pretrained CNN backbones used in the experiments and are consistent with the selected paper's methodology.

7.3 Data Augmentation

The augmentation pipeline included:

- random resized crop,
- horizontal flip,
- vertical flip,
- random rotation,
- color jitter.

These transformations were applied during training to improve robustness and reduce overfitting.

8 Exploratory Data Analysis

Exploratory data analysis was conducted to inspect the merged dataset structure and class composition. The analysis focused on class imbalance, per-class sample counts, and sample image inspection. The EDA confirmed that the dataset was not perfectly balanced and that certain disease categories had lower support than others. This imbalance justified the use of weighted evaluation metrics and motivated deeper class-wise analysis.

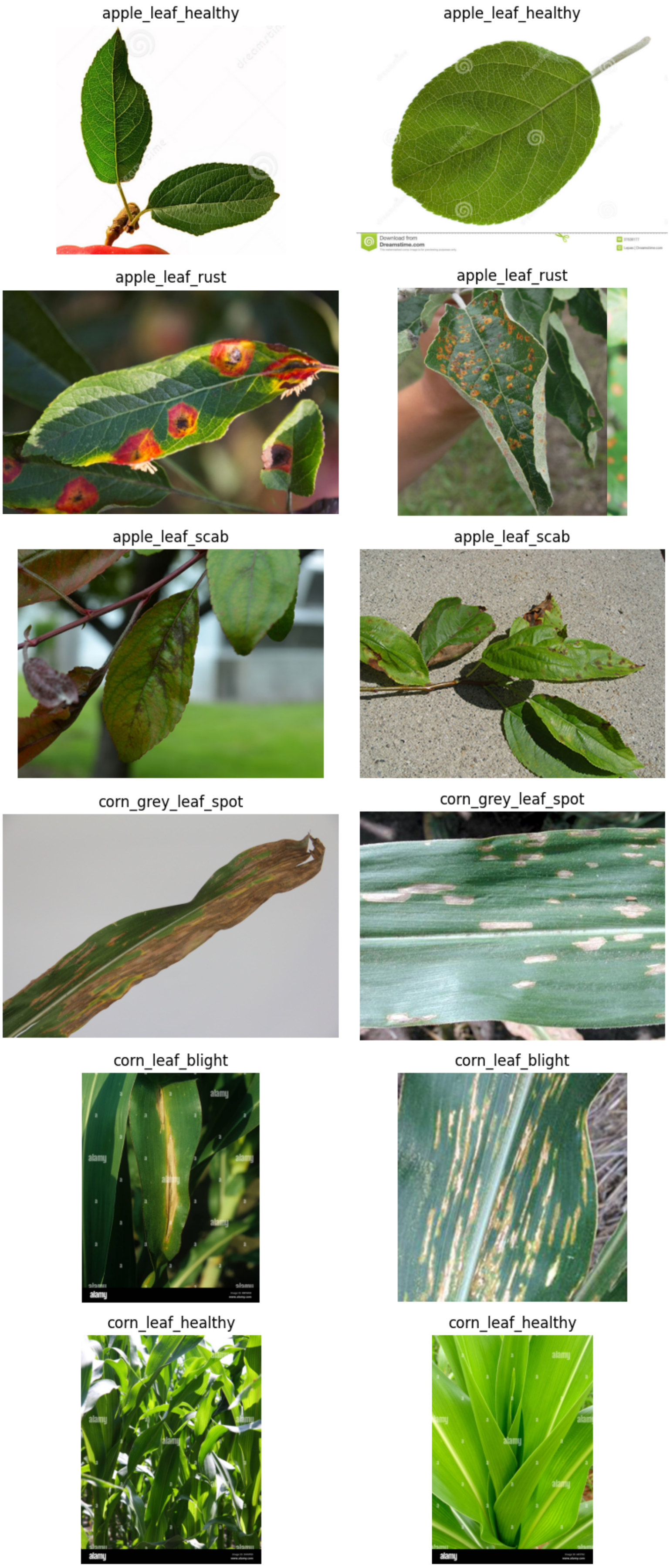

Figure 3: Representative sample images from multiple disease classes.

9 System Architecture and Implementation

The project was implemented in PyTorch. Six CNN architectures were evaluated:

- MobileNetV2
- ResNet18
- EfficientNet-B0
- ResNet50
- DenseNet201
- EfficientNet-B3

The selected paper explicitly evaluates ResNet50, DenseNet201, EfficientNet-B0, and EfficientNet-B3, while MobileNetV2 and ResNet18 were added as additional baselines.

Each architecture used:

- pretrained ImageNet weights,
- a modified classification head,
- dropout,
- a final fully connected output layer for 16 classes.

The model pipeline can be summarized as:

Input image → preprocessing → augmentation → CNN backbone → dropout → fully connected layer → disease prediction

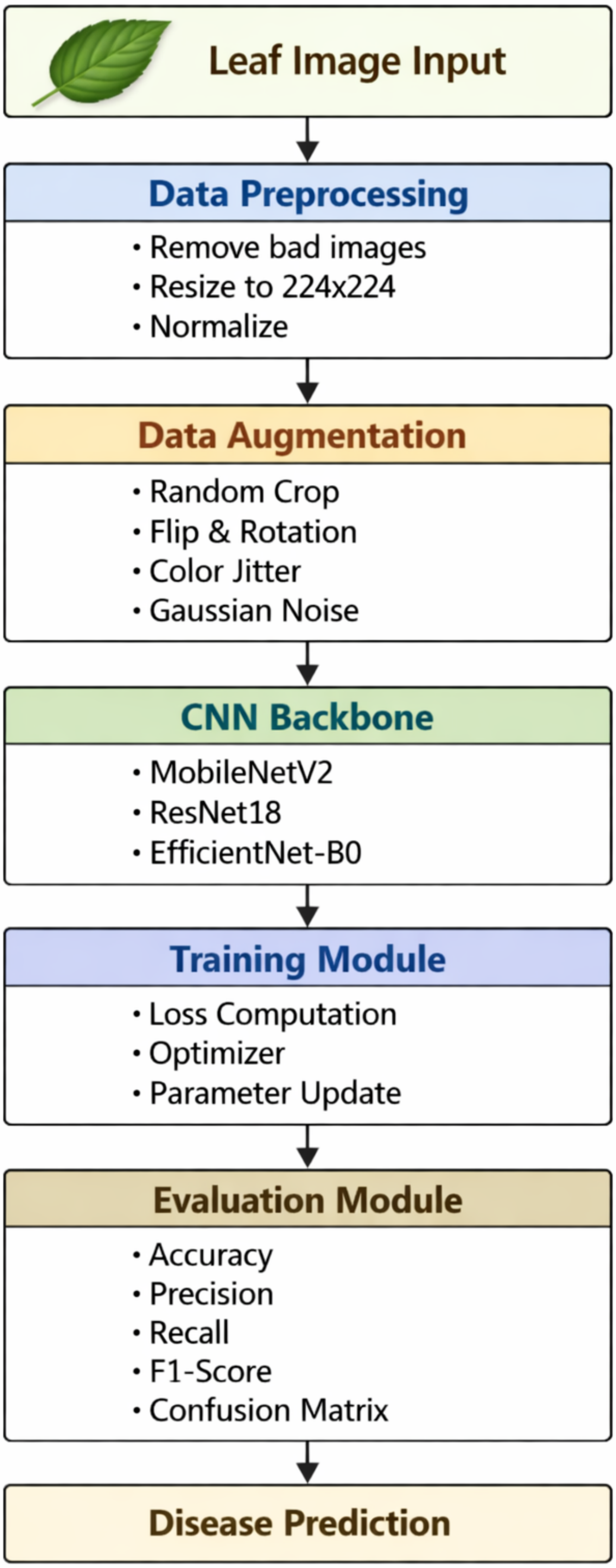

Figure 4: System architecture for multi-dataset crop disease detection.

10 Experimental Setup

The experiments were carried out under a common setup in order to make the comparison fair and reproducible.

10.1 Training Configuration

- Framework: PyTorch
- Image size: 224×224
- Optimizer: Adam
- Learning rate: 0.0001
- Dropout: 0.3
- Loss function: CrossEntropyLoss
- Learning rate scheduler: ReduceLROnPlateau

10.2 Hardware and Platform

Training and experimentation were carried out using local resources for debugging and Google Colab GPU resources for heavier model training. This allowed the project to scale from lightweight baseline experiments to the full comparison required for final evaluation.

11 Final Comparative Results

The final results for all six models are shown in Table 1.

Table 1: Comparative performance of all trained models.

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC
DenseNet201	0.7377	0.7491	0.7377	0.7360	0.9678
MobileNetV2	0.7260	0.7410	0.7260	0.7257	0.9666
ResNet50	0.7213	0.7381	0.7213	0.7207	0.9661
EfficientNet-B0	0.7190	0.7320	0.7190	0.7172	0.9635
EfficientNet-B3	0.7073	0.7092	0.7073	0.7019	0.9682
ResNet18	0.6956	0.7120	0.6956	0.6931	0.9624

The results show that DenseNet201 achieved the highest overall performance, while MobileNetV2 produced the second-best performance despite being a lightweight model. This is a significant practical result, since MobileNetV2 offers a strong trade-off between computational efficiency and predictive performance. ResNet50 and EfficientNet-B0 also performed competitively, while EfficientNet-B3 produced strong AUC-ROC but did not translate that advantage into the highest hard-classification accuracy. ResNet18 served as the weakest baseline but still produced useful reference performance.

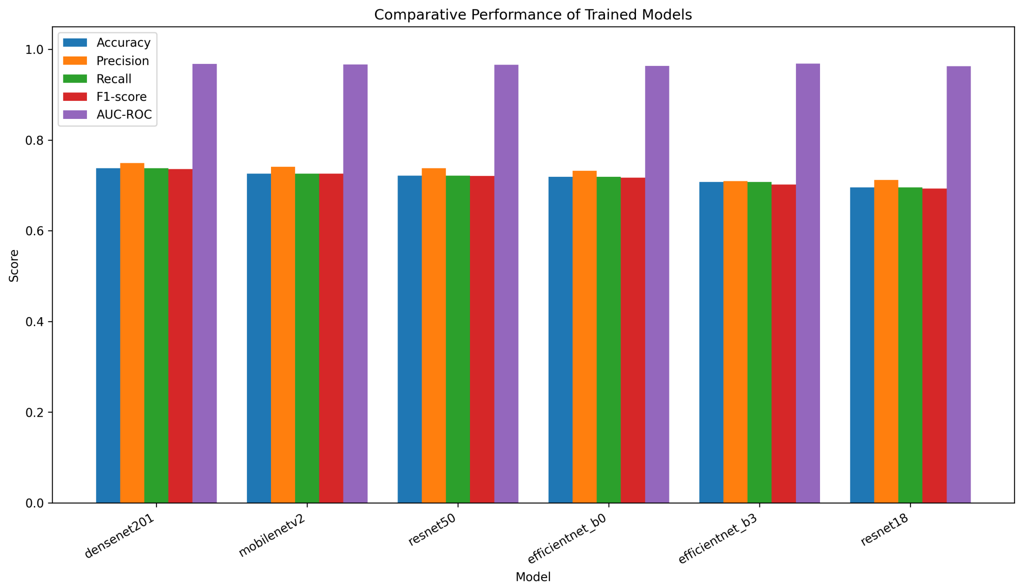

Figure 5: Final comparison of all trained models across accuracy and F1-score.

12 Model-Wise Discussion

12.1 DenseNet201

DenseNet201 achieved the best overall performance in this study, with an accuracy of 73.77% and weighted F1-score of 0.7360. Its strong performance suggests that dense connectivity is effective for crop disease recognition because it supports feature reuse across layers and helps preserve fine-grained visual information. In plant disease detection, this is important because disease symptoms often appear as subtle texture, lesion, or color variations.

DenseNet201 performed particularly well on apple_leaf_healthy, apple_leaf_rust, corn_leaf_rust, tomato_leaf_mould, and tomato_leaf_healthy. Its weakest classes were tomato_leaf_bacterial_spot and tomato_leaf_early_blight, which remained difficult across nearly all models.

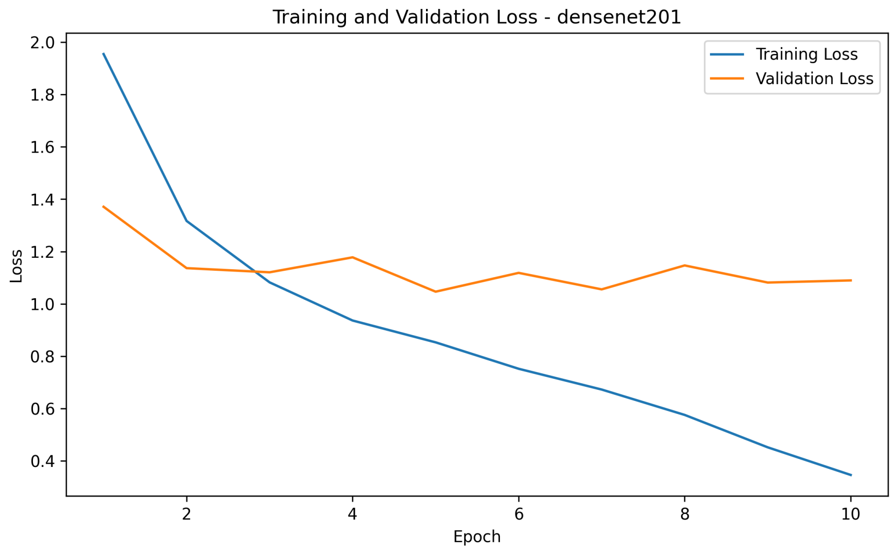

Figure 6: DenseNet201 training and validation loss curve.

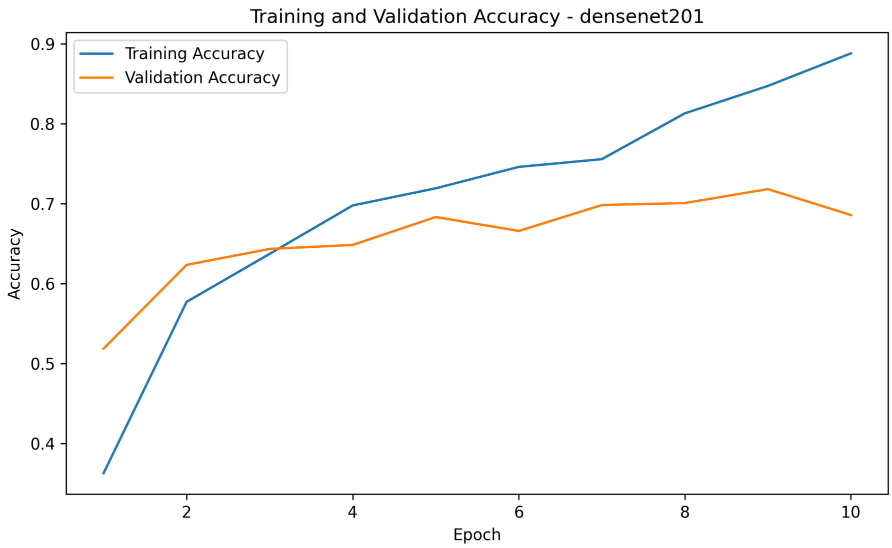

Figure 7: DenseNet201 training and validation accuracy curve.

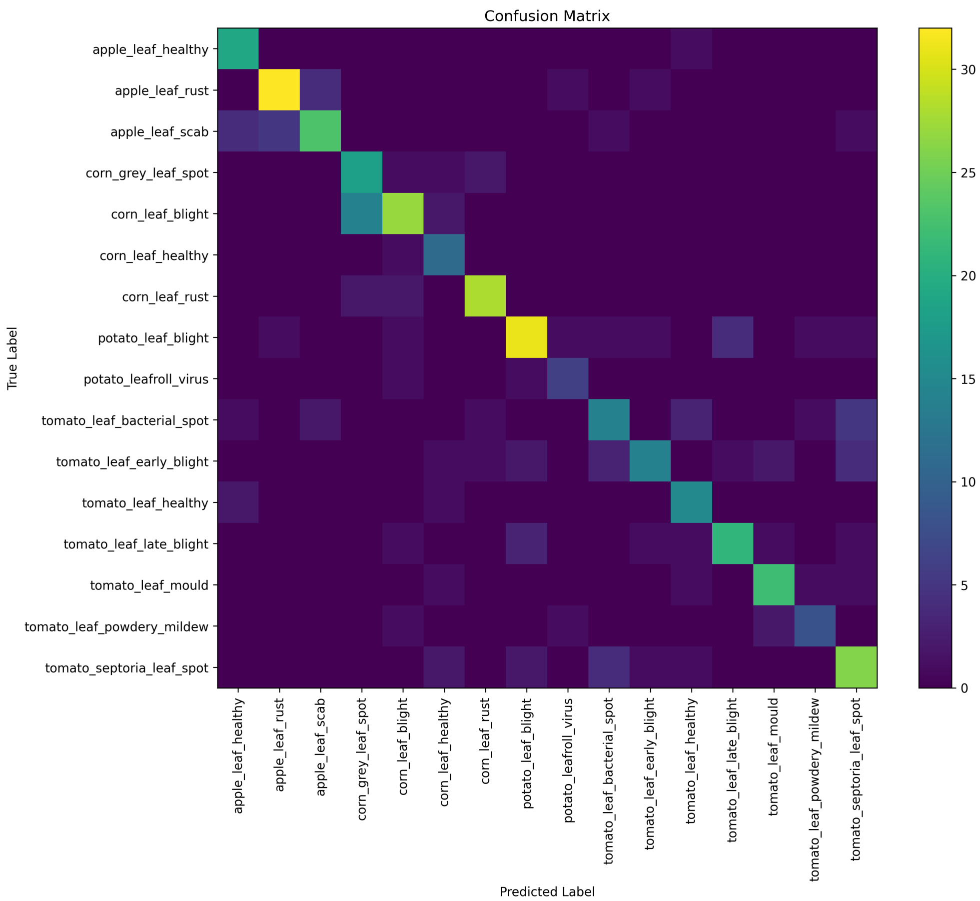

Figure 8: DenseNet201 confusion matrix.

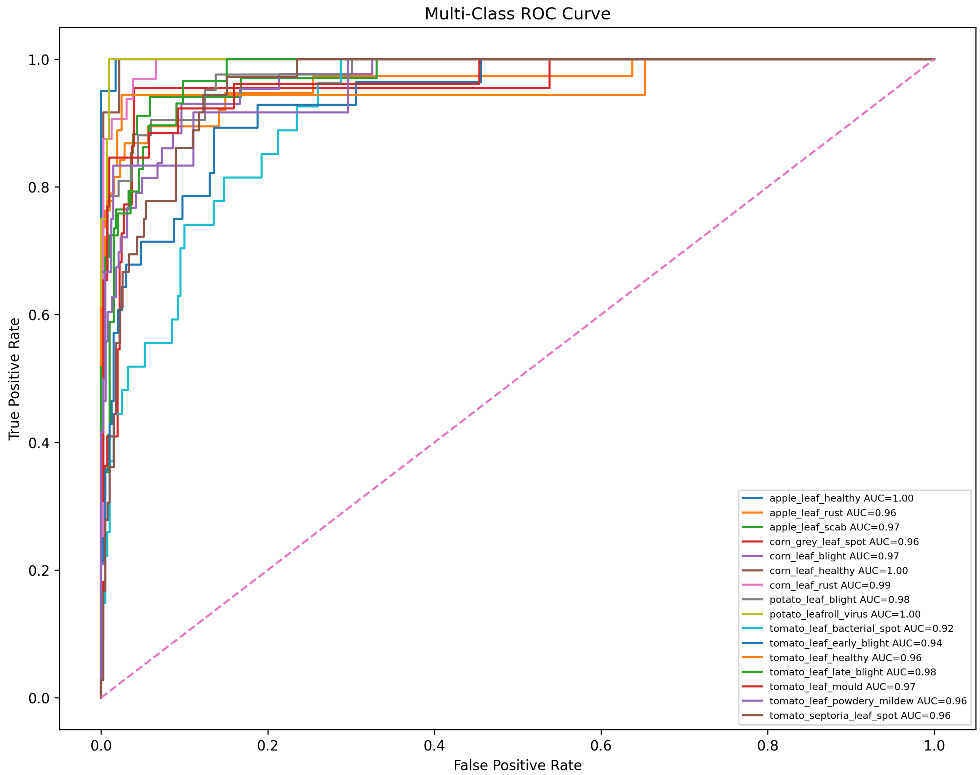

Figure 9: DenseNet201 ROC curve.

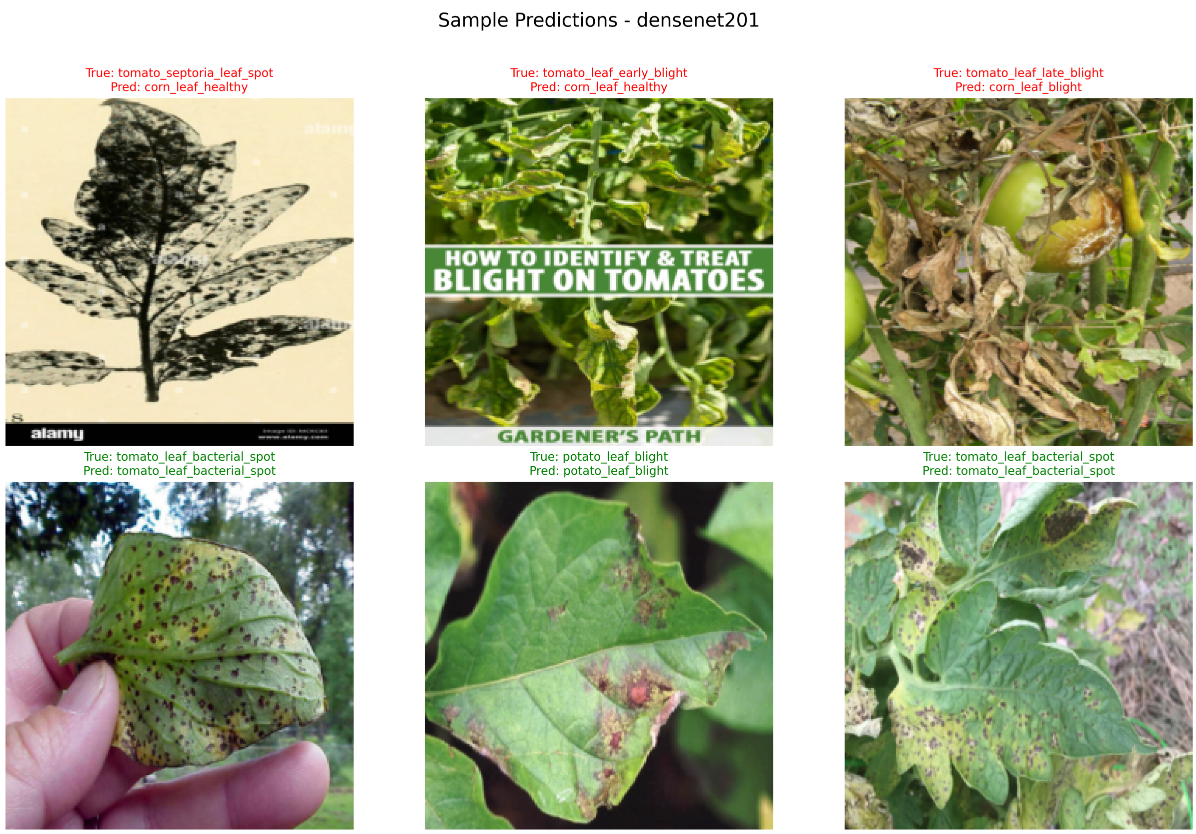

Figure 10: DenseNet201 sample predictions on test set images, showing predicted and true labels.

12.2 MobileNetV2

MobileNetV2 produced one of the most important outcomes in the project. Despite being a lightweight model, it achieved the second-best overall performance with an accuracy of 72.60% and weighted F1-score of 0.7257. This result demonstrates that lightweight architectures can remain highly competitive when trained on a diverse multi-dataset image collection.

MobileNetV2 performed strongly on apple_leaf_healthy, apple_leaf_rust, corn_leaf_rust, potato_leaf_blight, tomato_leaf_healthy, and tomato_leaf_mould. Its weakest categories were tomato_leaf_bacterial_spot, tomato_leaf_early_blight, and potato_leafroll_virus. The strength of MobileNetV2 makes it particularly attractive for eventual practical deployment.

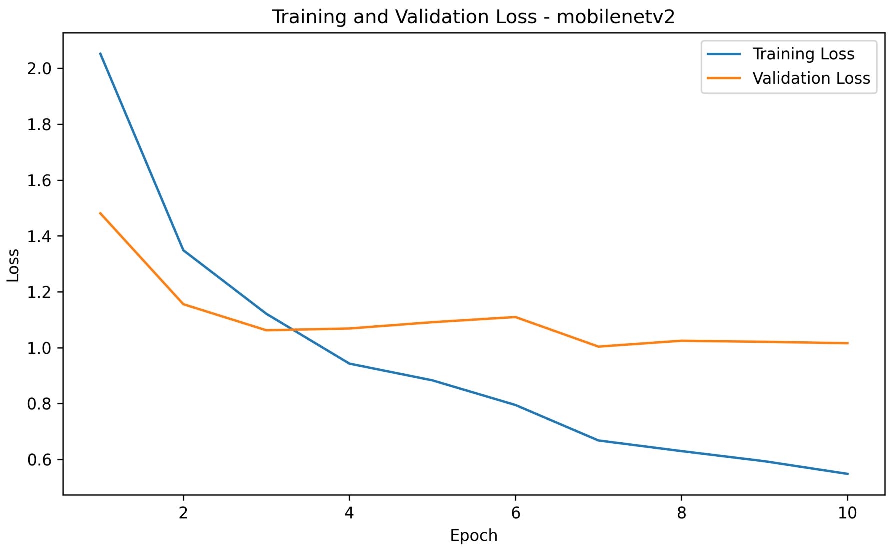

Figure 11: MobileNetV2 training and validation loss curve.

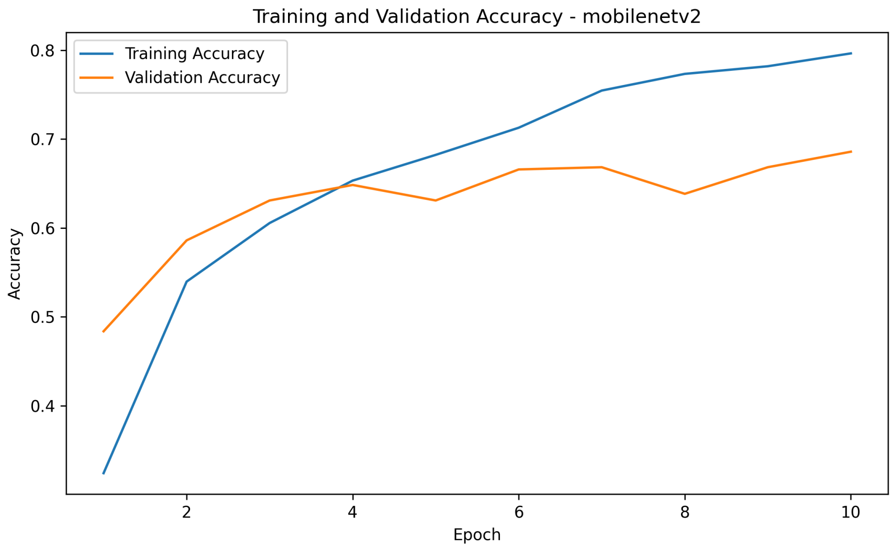

Figure 12: MobileNetV2 training and validation accuracy curve.

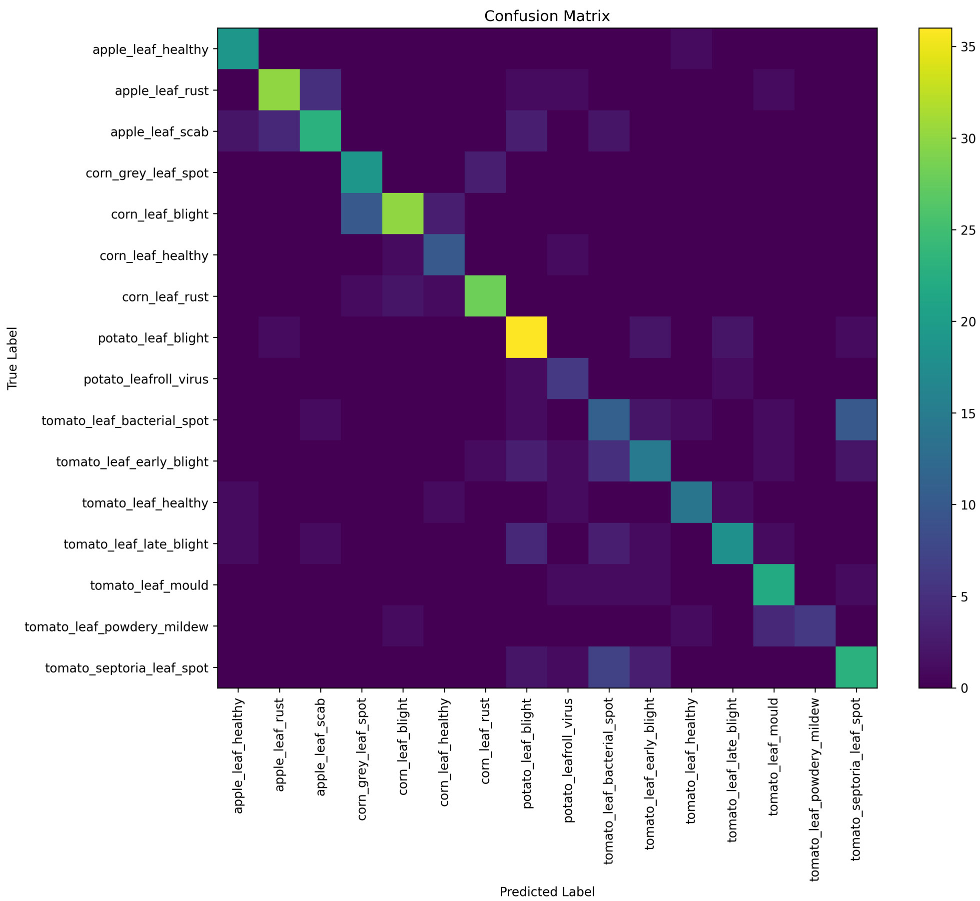

Figure 13: MobileNetV2 confusion matrix.

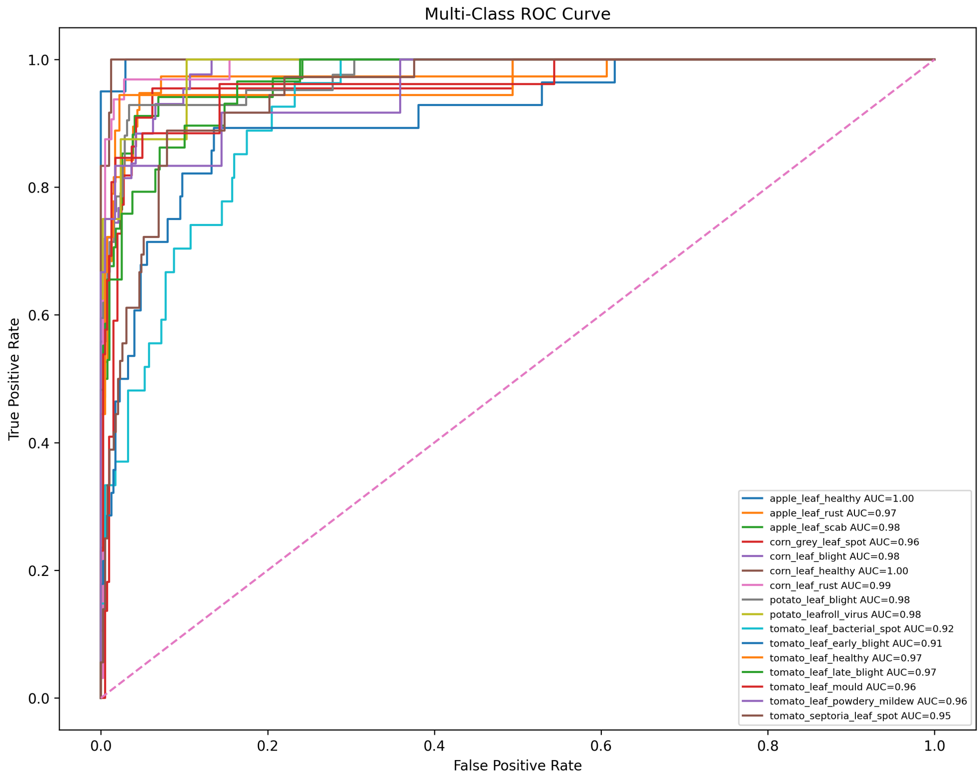

Figure 14: MobileNetV2 ROC curve.

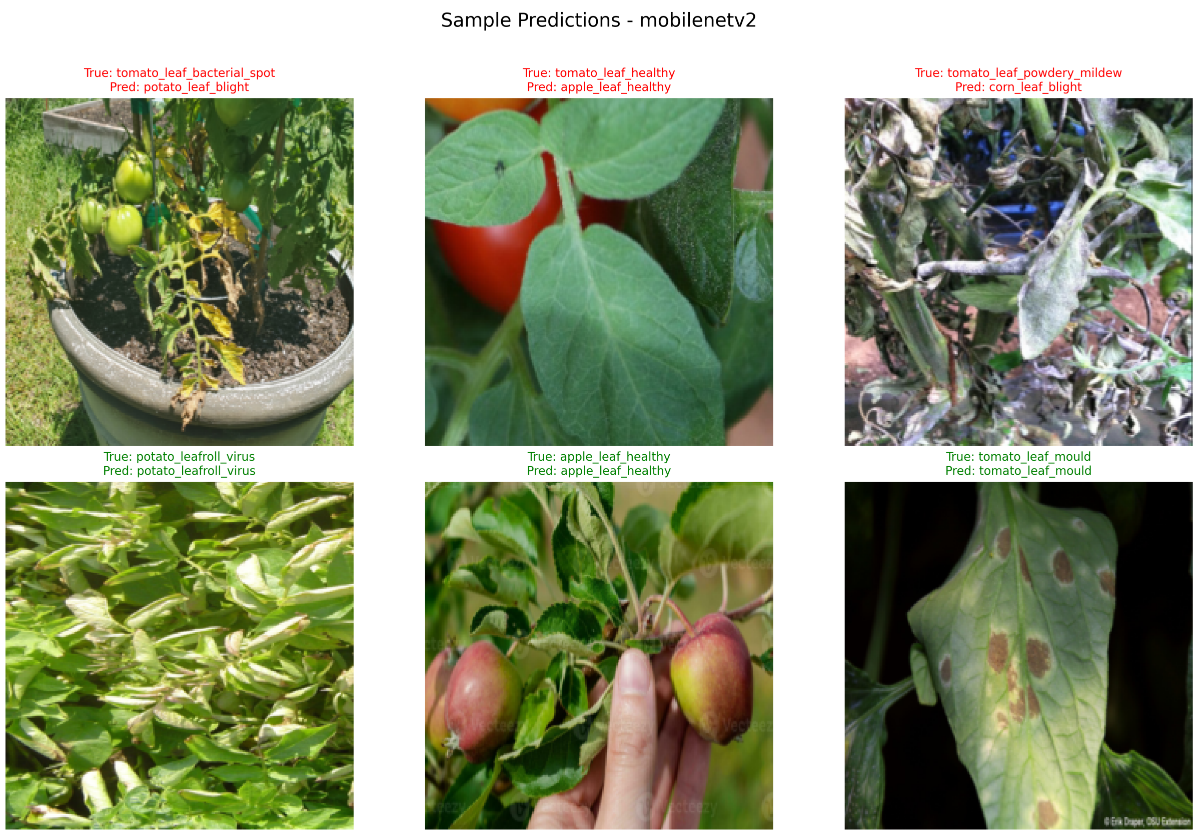

Figure 15: MobileNetV2 sample predictions on test set images, showing predicted and true labels.

12.3 ResNet50

ResNet50 ranked third overall with an accuracy of 72.13% and weighted F1-score of 0.7207. It performed strongly on several classes, including apple_leaf_healthy, apple_leaf_scab, corn_leaf_rust, and tomato_leaf_late_blight. However, it struggled on tomato_leaf_bacterial_spot and tomato_leaf_early_blight.

Although ResNet50 is a powerful architecture, it did not surpass DenseNet201 or MobileNetV2 in this setting. This suggests that residual depth alone was not sufficient to maximize performance under the current dataset and training setup.

Figure 16: ResNet50 training and validation loss curve.

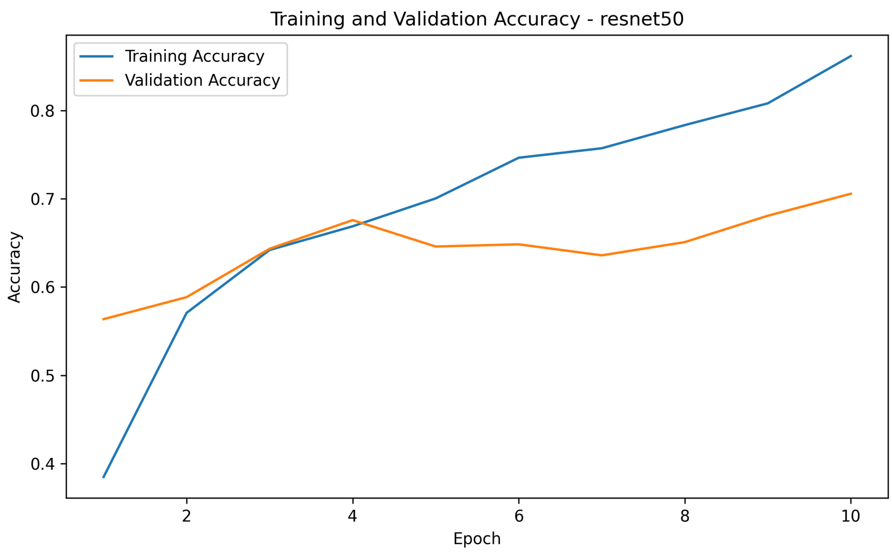

Figure 17: ResNet50 training and validation accuracy curve.

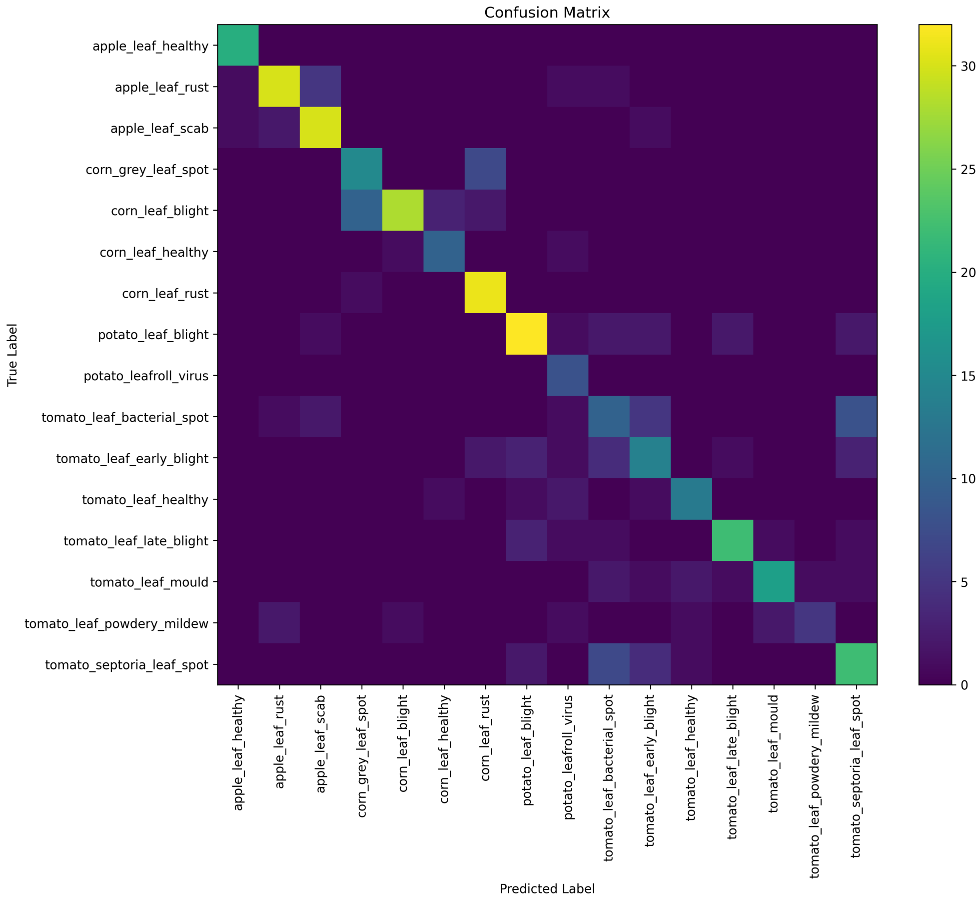

Figure 18: ResNet50 confusion matrix.

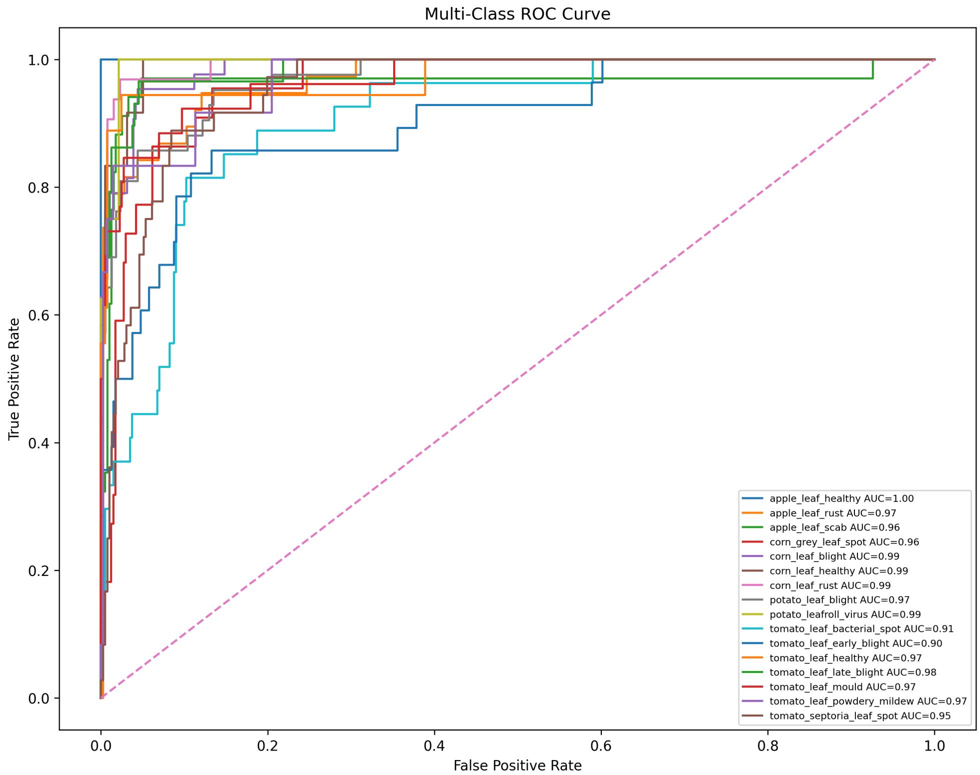

Figure 19: ResNet50 ROC curve.

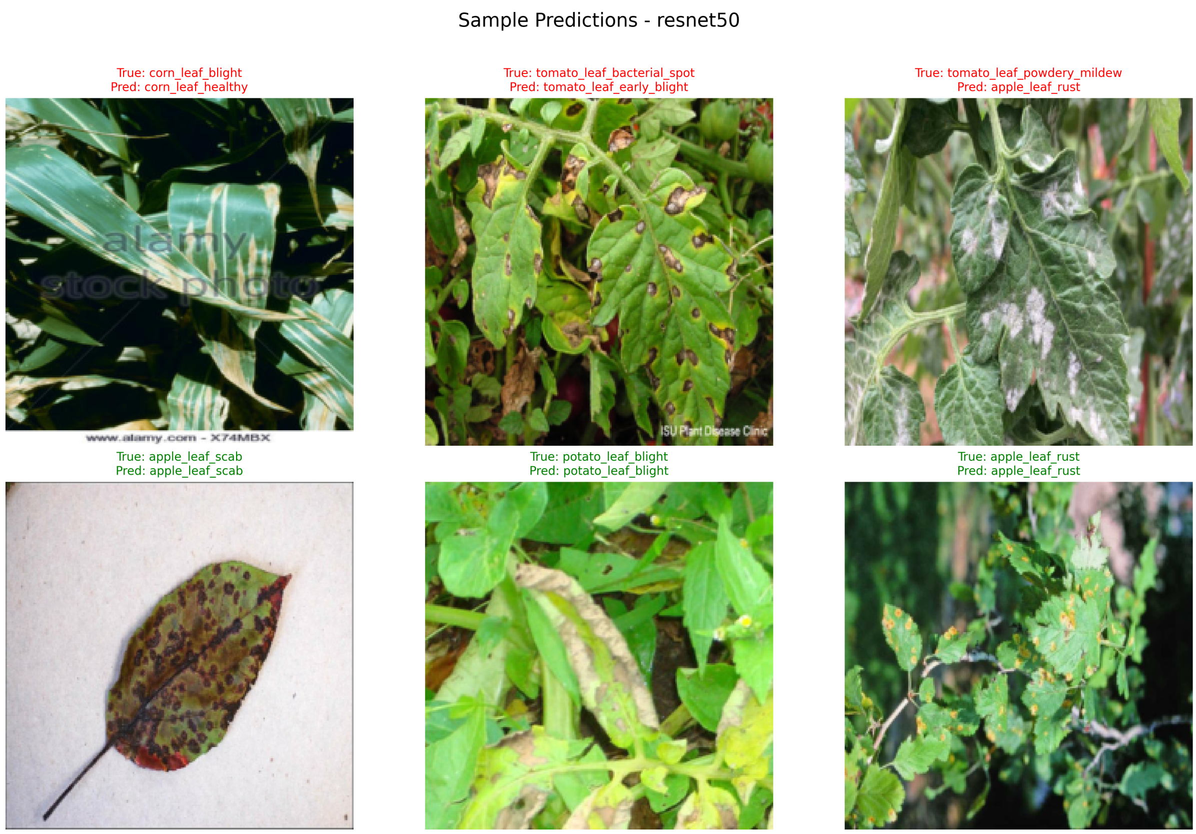

Figure 20: ResNet50 sample predictions on test set images, showing predicted and true labels.

12.4 EfficientNet-B0

EfficientNet-B0 was one of the main paper-aligned models and achieved an accuracy of 71.90% with weighted F1-score of 0.7172. It performed strongly on apple_leaf_healthy, potato_leaf_blight, tomato_leaf_healthy, tomato_leaf_late_blight, and tomato_leaf_powdery_mildew.

Although EfficientNet-B0 did not emerge as the strongest overall model, it delivered stable and competitive performance across many classes and remained one of the more balanced architectures in the experiment.

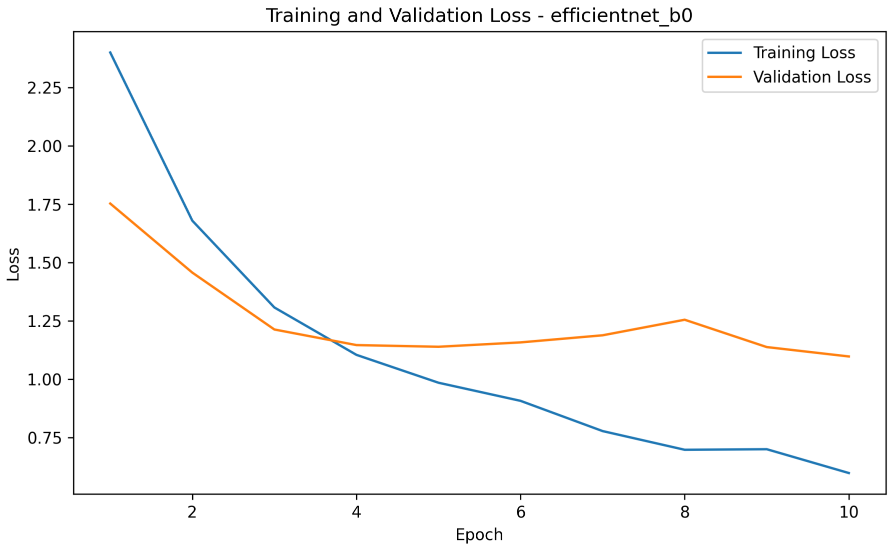

Figure 21: EfficientNet-B0 training and validation loss curve.

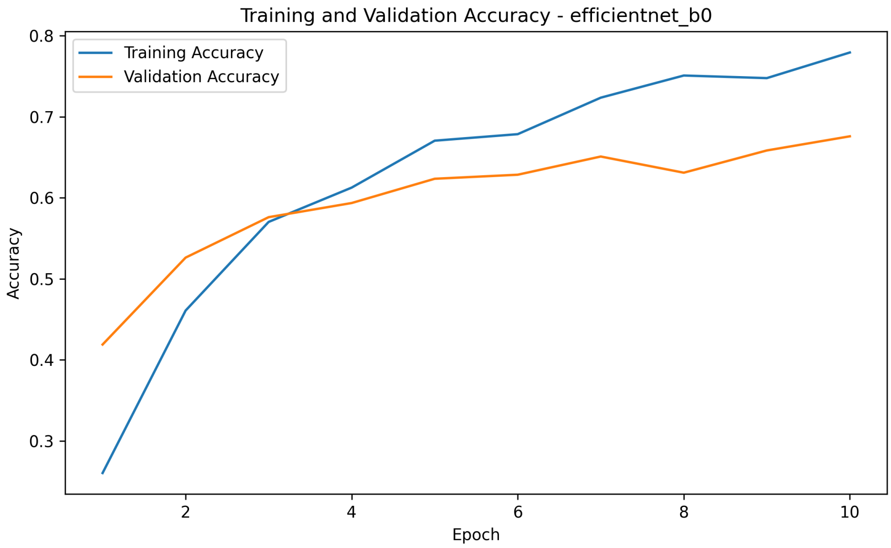

Figure 22: EfficientNet-B0 training and validation accuracy curve.

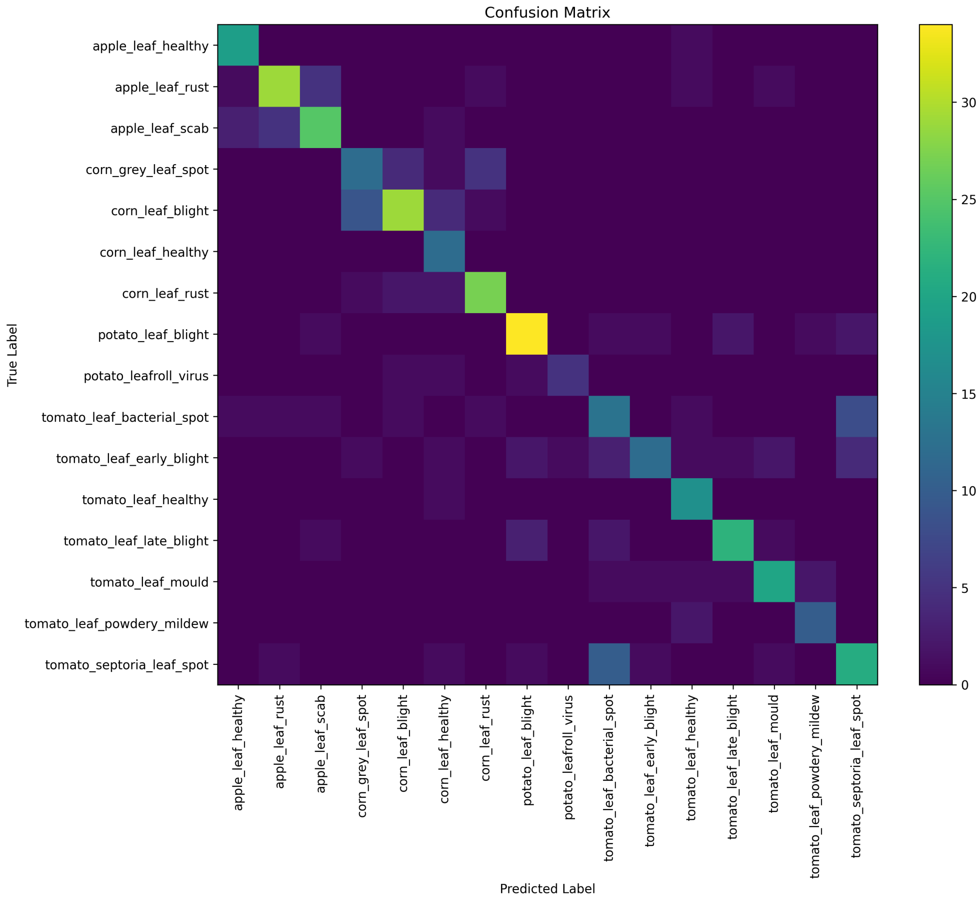

Figure 23: EfficientNet-B0 confusion matrix.

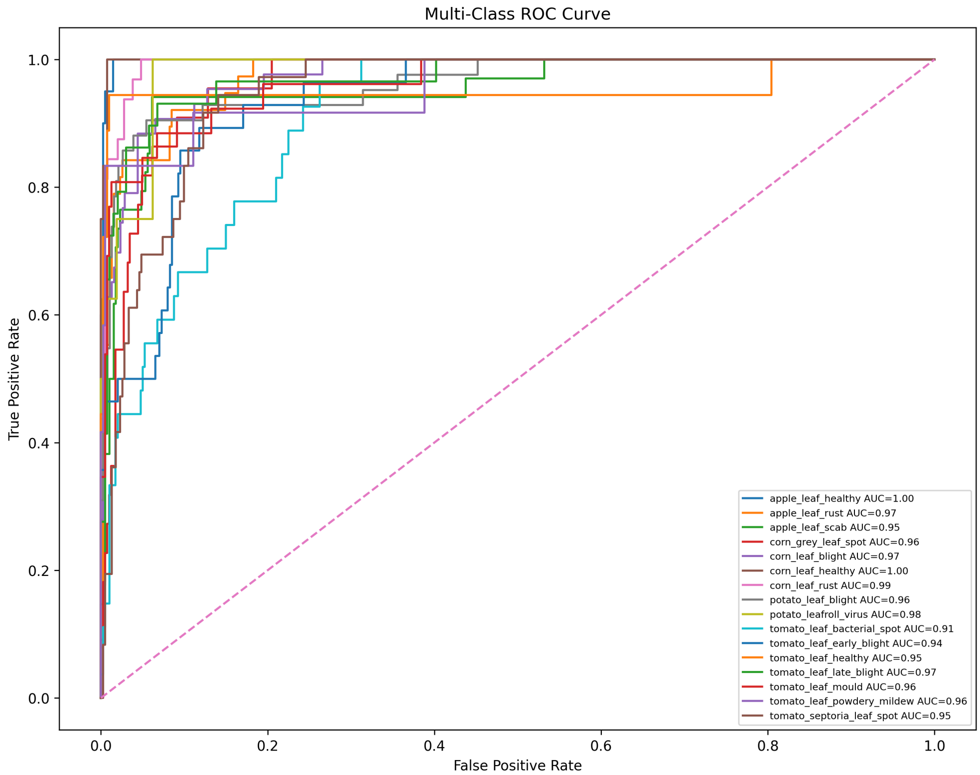

Figure 24: EfficientNet-B0 ROC curve.

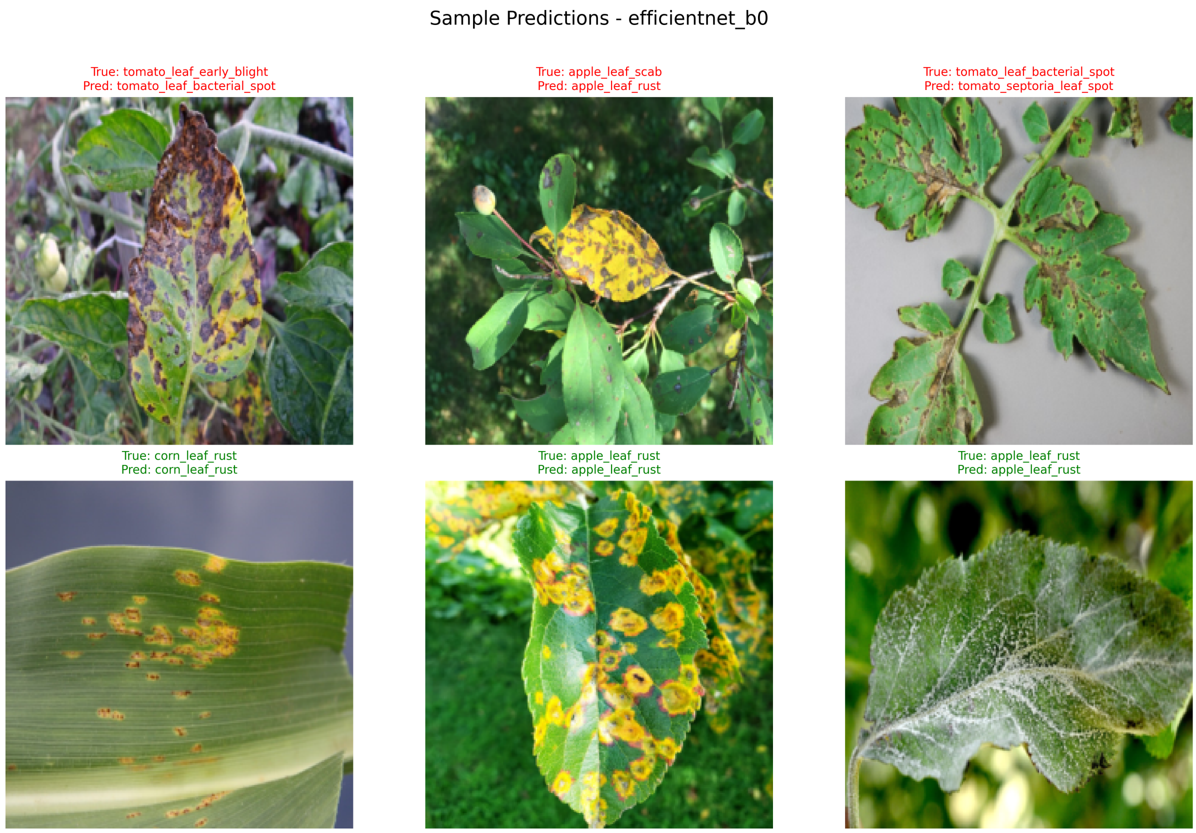

Figure 25: EfficientNet-B0 sample predictions on test set images, showing predicted and true labels.

12.5 EfficientNet-B3

EfficientNet-B3 achieved an accuracy of 70.73% and weighted F1-score of 0.7019. While it did not outperform EfficientNet-B0, it produced a very strong AUC-ROC of 0.9682. This indicates that it ranked class probabilities well, even though final hard-label classification remained less effective than expected.

Its strongest classes included corn_leaf_healthy, corn_leaf_rust, apple_leaf_healthy, and tomato_leaf_powdery_mildew. Its weakest categories were tomato_leaf_bacterial_spot, tomato_leaf_early_ and tomato_septoria_leaf_spot. This suggests that EfficientNet-B3 may require more careful tuning or more training data to fully realize its capacity.

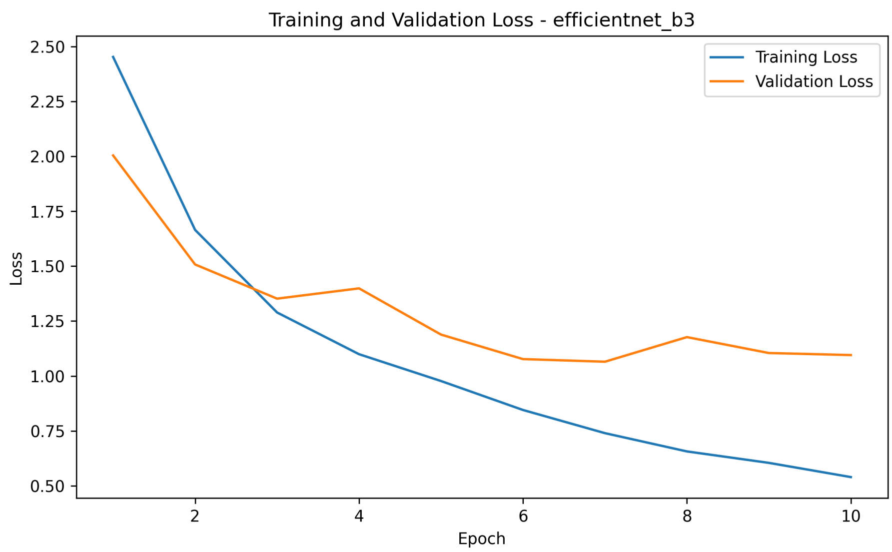

Figure 26: EfficientNet-B3 training and validation loss curve.

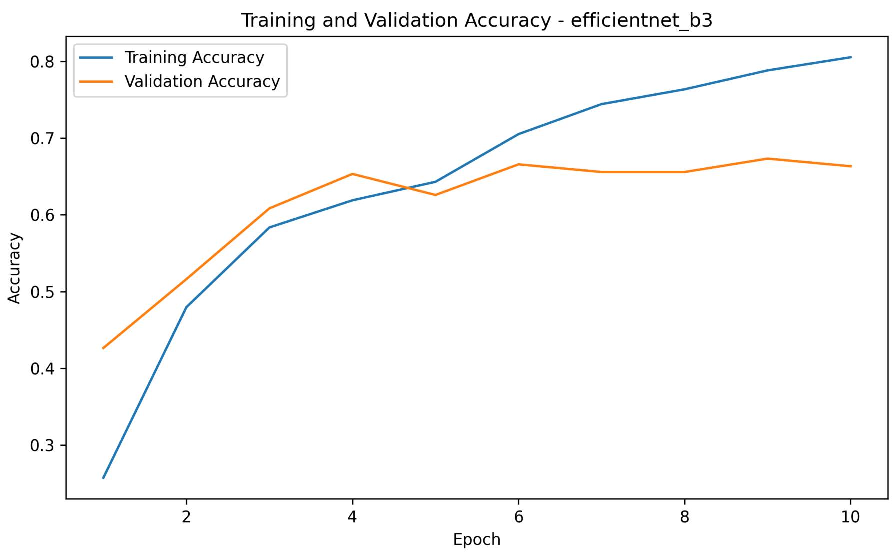

Figure 27: EfficientNet-B3 training and validation accuracy curve.

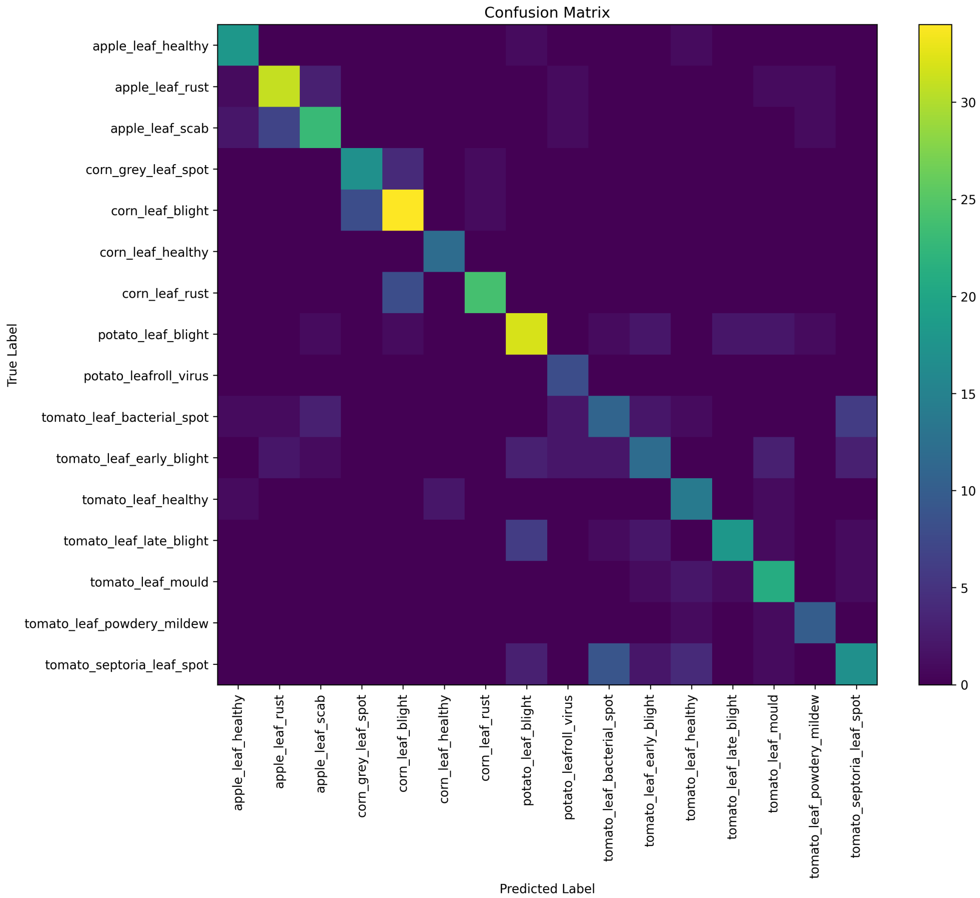

Figure 28: EfficientNet-B3 confusion matrix.

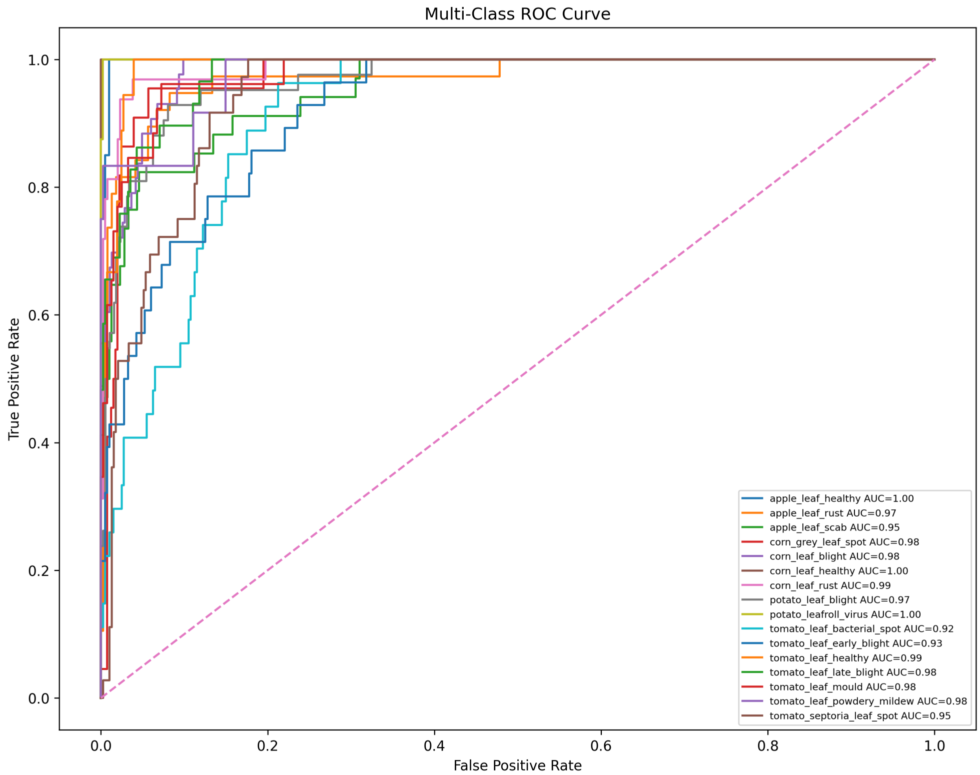

Figure 29: EfficientNet-B3 ROC curve.

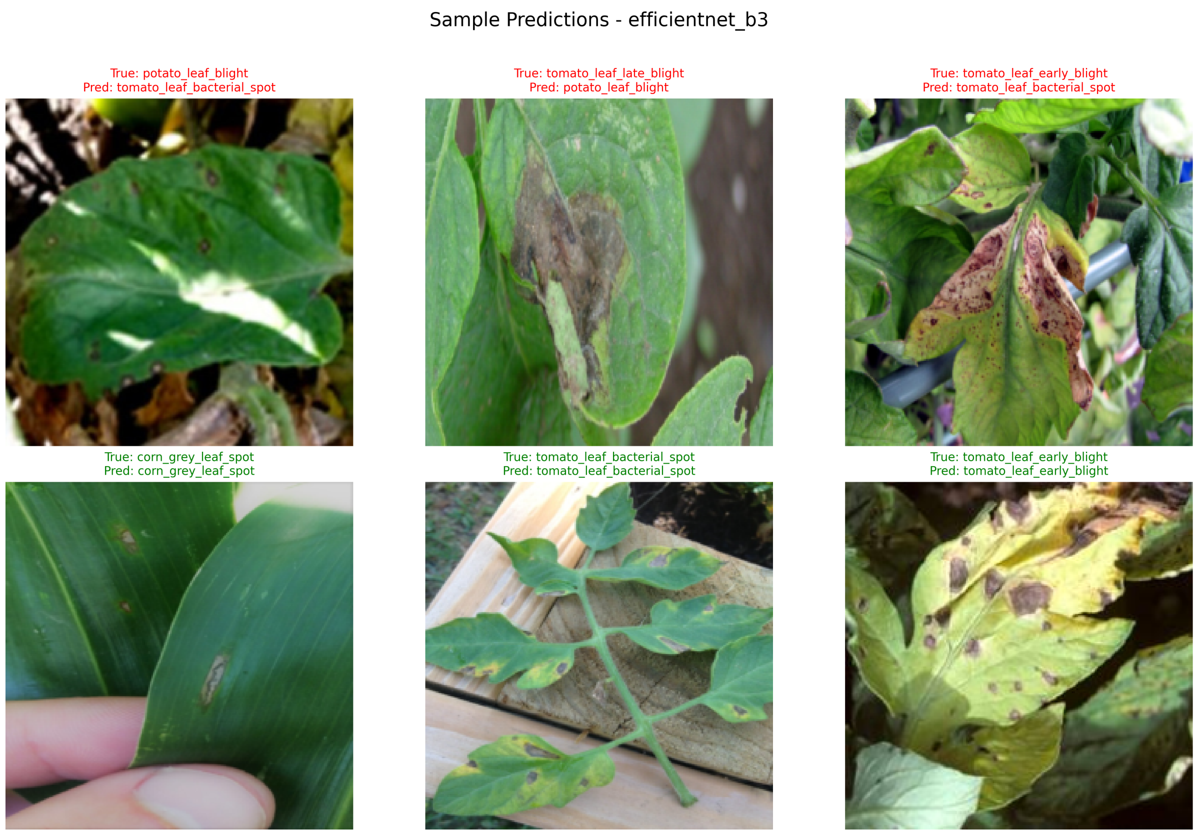

Figure 30: EfficientNet-B3 sample predictions on test set images, showing predicted and true labels.

12.6 ResNet18

ResNet18 served as a baseline model and produced the lowest overall performance, with an accuracy of 69.56% and weighted F1-score of 0.6931. Despite being the weakest model, it still produced useful baseline behavior and validated the end-to-end implementation pipeline.

ResNet18 performed relatively well on apple_leaf_healthy, corn_leaf_rust, tomato_leaf_powdery_mildev and potato_leaf_blight. However, it struggled on tomato_septoria_leaf_spot, tomato_leaf_early_blight, and tomato_leaf_bacterial_spot. Its lower capacity may have limited its ability to distinguish fine-grained disease features.

Figure 31: ResNet18 training and validation loss curve.

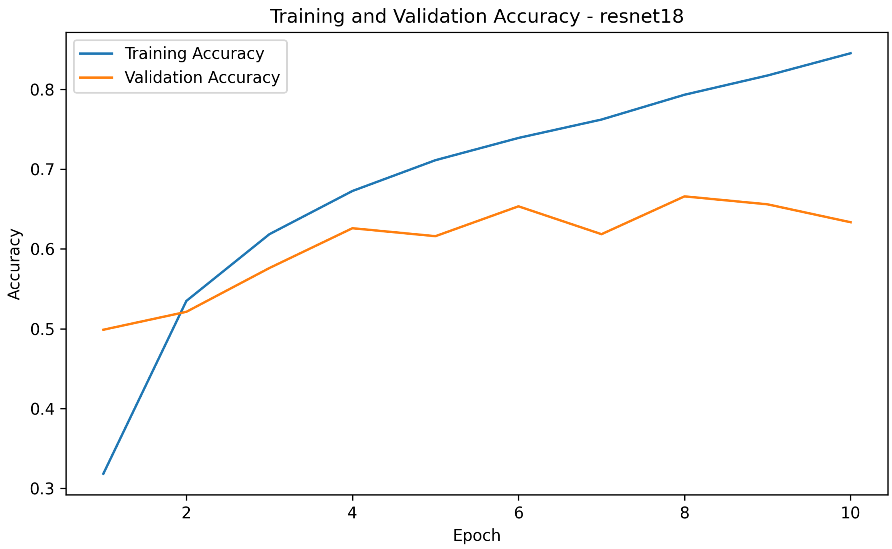

Figure 32: ResNet18 training and validation accuracy curve.

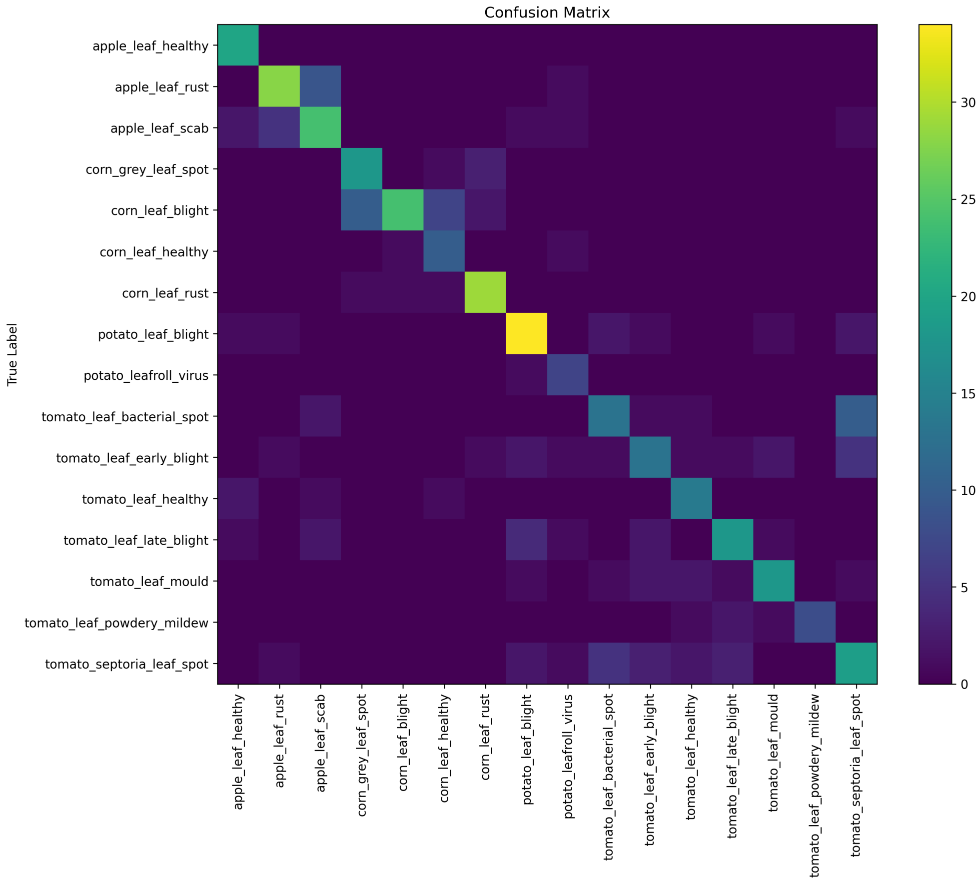

Figure 33: ResNet18 confusion matrix.

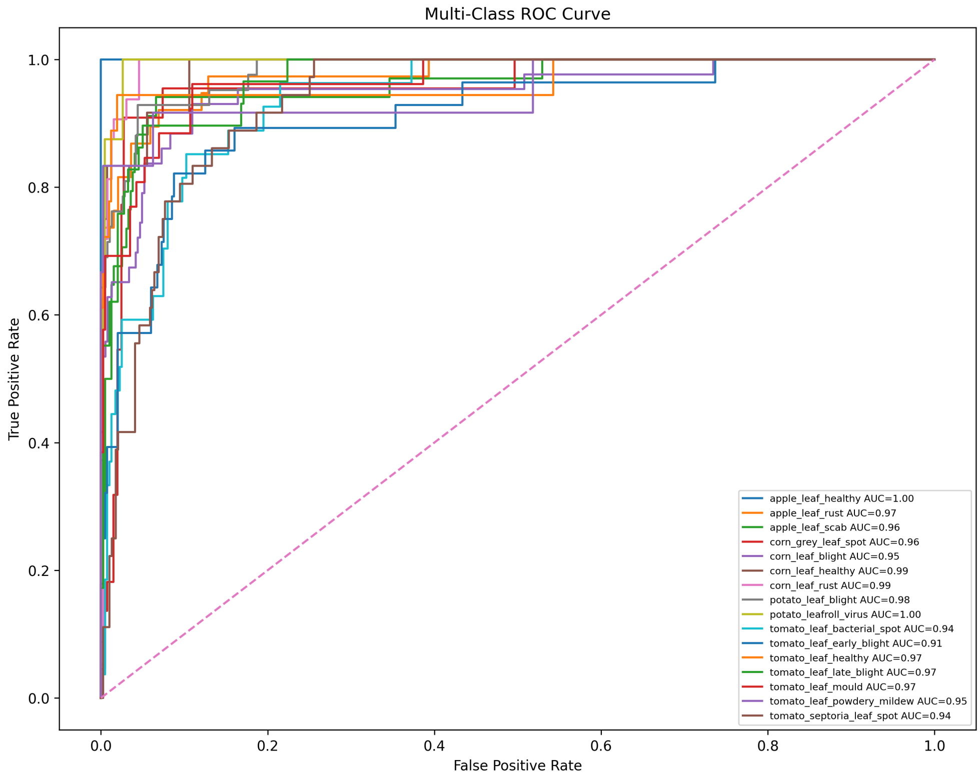

Figure 34: ResNet18 ROC curve.

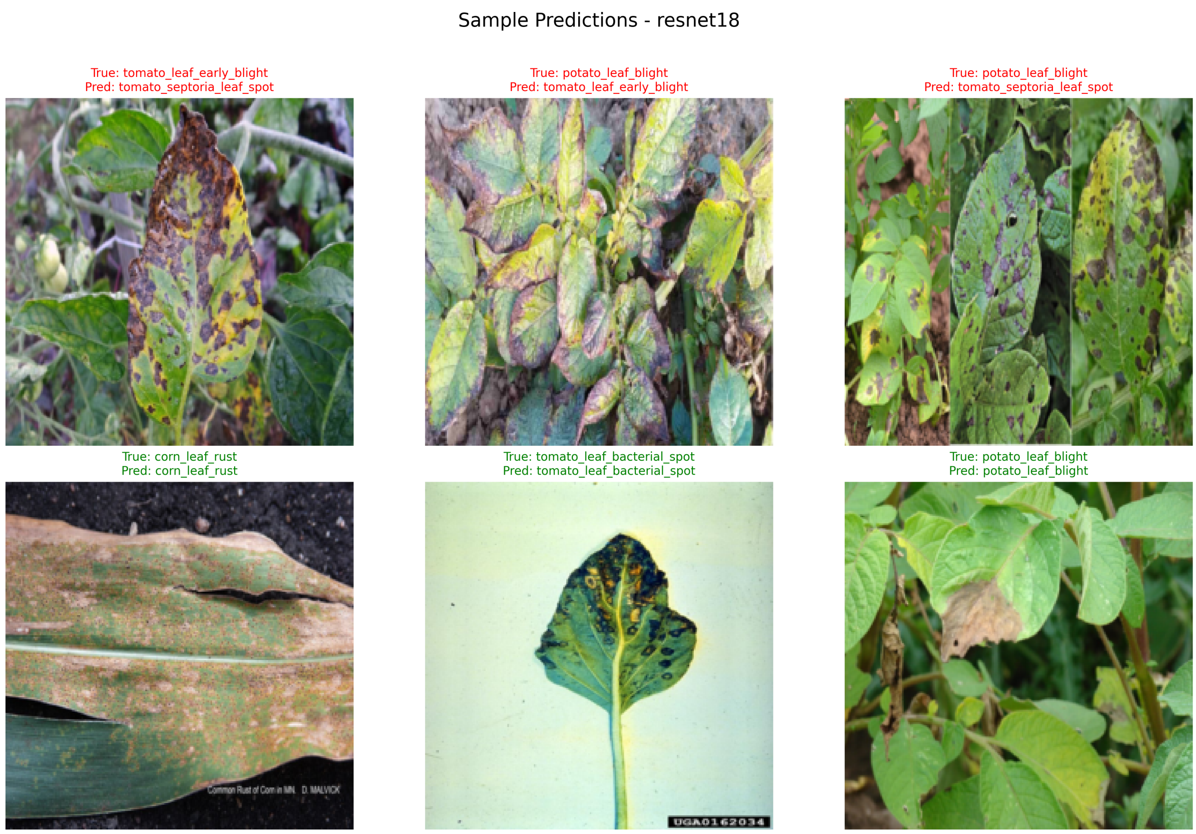

Figure 35: ResNet18 sample predictions on test set images, showing predicted and true labels.

13 Detailed Comparative Discussion

The comparative findings reveal several important patterns.

First, larger models were not automatically better. DenseNet201 was the best model, but EfficientNet-B3 did not outperform EfficientNet-B0, and ResNet50 did not surpass DenseNet201. This indicates that architecture design, feature reuse, optimization behavior, and dataset compatibility matter just as much as model scale.

Second, MobileNetV2 achieved a very strong result despite being lightweight. This is one of the most practically important outcomes of the project. It suggests that efficient architectures can still deliver high-quality crop disease recognition, making them attractive candidates for future deployment-oriented systems.

Third, the class-wise results show that all models struggled most with fine-grained tomato disease categories. These diseases share overlapping visible symptoms such as spots, lesions, discoloration, and irregular tissue damage. The repeated difficulty across all architectures suggests that this is a dataset and task challenge, not merely a model-selection issue.

Fourth, AUC-ROC scores were consistently high across all models, even when accuracy and F1-score differed. This indicates that several architectures were capable of learning useful probability distributions, but the final hard decision boundary remained imperfect for difficult classes.

Overall, the best performing model, DenseNet201, appears to have benefited from dense feature reuse, while MobileNetV2 benefited from an efficient design that generalized well despite lower computational complexity.

14 Error Analysis

The final results reveal clear error patterns. The hardest categories across most models were tomato_leaf_bacterial_spot, tomato_leaf_early_blight, and tomato_septoria_leaf_spot. These classes often produce similar lesion morphology and overlapping color changes, which reduces class separability.

Another recurring issue was lower support for certain classes, such as potato_leafroll_virus. Smaller support makes performance more unstable and more sensitive to small misclassification counts.

In contrast, visually distinctive classes such as apple_leaf_healthy, corn_leaf_rust, and tomato_leaf_mould were more consistently classified across models.

The error analysis therefore suggests that the remaining challenge is not simply model strength, but also fine-grained disease similarity and the need for more balanced and class-diverse data.

15 Bias Check

The primary form of bias in this system is dataset bias rather than demographic bias. The main sources are:

- class imbalance,
- source-domain variation between PlantDoc and web-sourced data,
- unequal visual distinctiveness between disease categories.

Some classes are inherently easier because their symptoms are more distinctive, while others are harder because they share overlapping visual cues. In addition, multi-dataset training improves robustness but also introduces the possibility that a model may learn source-specific image characteristics.

These biases were partially addressed through augmentation, consistent preprocessing, and weighted evaluation metrics, but they were not completely eliminated.

16 Limitations

This implementation has several limitations:

1. The dataset remains imbalanced across classes.
2. Visually similar tomato disease classes remain difficult.
3. The implementation uses multi-dataset training but does not include an explicit domain adaptation module.
4. Some models may require more tuning or longer training to fully realize their potential.
5. Although the system is technically strong, it is still an academic implementation rather than a production agricultural platform.

17 GitHub Collaboration and Commit History

The GitHub repository was maintained through command-line version control and collaborative commits from both team members. The repository was progressively updated across preprocessing, model development, training, evaluation, result analysis, and final cleanup.

17.1 Contributor Profiles

- Syed Muhammad Ubaid Ur Rehman – https://github.com/SyedMuhammadUbaidUrRehman22
- Hafiz Salman Saeed – https://github.com/HMSalmanSaeed

17.2 Commit History Evidence

The project commit history shows active contributions from both collaborators across multiple stages of the workflow. Important commits included preprocessing revisions, metrics and evaluation modules, model factory creation, training script updates, model-specific updates, folder structure improvements, and final result integration. Notable commit activity included items such as:

- "Revised Preprocessing module"
- "Add metrics module and Evaluate module"
- "Model factory and Training the model"
- "Update train.py"
- "Model Update"
- "Revised CDD_upload"
- "Final Update (Training done on colab)"
- "Analysis and Model comparison"

17.3 Contribution Summary

Contributor	GitHub Username	Contribution Summary
Syed Muhammad Ubaid Ur Rehman	Subaid04	Dataset setup, merge and split pipeline, notebook integration, training workflow organization, model experimentation, output analysis, final comparison, final report integration, repository cleanup
Hafiz Salman Saeed	HMSalmanSaeed	Preprocessing refinement, metrics and evaluation module work, folder structure clarity, training script update, baseline experimentation, collaborative code revision

18 Conclusion

This project successfully implemented the paper *Plant Leaf Disease Detection Using Deep Learning: A Multi-Dataset Approach* as a Track C research paper implementation. A complete technical pipeline was developed, covering data collection, cleaning, preprocessing, EDA, model training, evaluation, comparative analysis, and interpretation.

The results show that DenseNet201 was the strongest overall architecture for this task, while MobileNetV2 provided a highly competitive and efficient alternative. The experiments also demonstrated that larger models do not automatically guarantee better performance, as shown by the relative underperformance of EfficientNet-B3 compared with DenseNet201 and MobileNetV2. The most difficult categories across all models were the visually similar tomato disease classes, which confirms that fine-grained symptom similarity remains a central challenge in crop disease classification.

The findings support the selected paper's core argument that multi-dataset training improves robustness and makes experiments more realistic under diverse image conditions. However, class imbalance, domain shift, and subtle inter-class similarity continue to limit performance. Future work should focus on more balanced data collection, stronger domain generalization strategies, and possible deployment of a lightweight user-facing diagnostic interface.

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